

# Intro to Bayes Notes

M08: Lecture 1

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## Introduction

### Notation

- $\theta$  = unobserved vector quantity or pop parameter of interest
- $y$  = observed data
- $\tilde{y}$  or  $y_{n+1}$  unknown but potentially observable quantities

### Bayesian inference

- Probability statements conditional on  $y$ 
  - $p(\theta|y)$  or  $p(\tilde{y}|y)$
  - Also on known values of any covariates

### Bayes' rule

- To make probability statements about  $\theta|y$ , start with joint probability distribution
  - Can be written as product of two densities that are referred to as the prior distribution and the sampling distribution
  - $p(\theta, y) = p(\theta)p(y|\theta)$

- Conditioning on the known value of the data, using Bayes' rule we get the posterior density
  - $p(\theta|y) = \frac{p(\theta,y)}{p(y)} = \frac{p(\theta)p(y|\theta)}{p(y)}$ 
    - \* Where  $p(y) = \sum_{\theta} p(\theta)p(y|\theta)$ 
      - Or the integral for continuous  $\theta$
- We can generalize this to functions
  - Let
    - \*  $y = (y_1, \dots, y_n)$
    - \*  $f(y|\theta)$  be the model for the observed data (usually a probability distribution)
    - \*  $\pi(\theta)$  be the prior distribution for  $\theta$
  - The posterior distribution for inference on  $\theta$ :
    - \*  $f(\theta|y) = \frac{f(y|\theta)\pi(\theta)}{\int f(y|u)\pi(u)du}$ 
      - $\int f(y|u)\pi(u)du$  is the marginal density of the data  $y$  and doesn't depend on  $\theta$ !
      - The integral can be very difficult to evaluate
      - When the data is fixed, the integral can be regarded as a normalizing constant.

## MCMC Methods

- MCMC allows us to avoid evaluating the integral (marginal density)
- Makes use of the unnormalized posterior density
  - $f(\theta|y) \propto f(y|\theta)\pi(\theta)$
- We can rewrite this in terms of the likelihood function
  - $f(\theta|y) \propto \mathcal{L}(\theta)\pi(\theta)$
- MCMC allows for easier evaluation of the likelihood when you have unobserved data
  - Augments the data to make estimation simpler

## Motivating example

### Chain Binomial

- Assume we have a Reed-Frost chain binomial model
  - Independent households where there is a single index case in each household
  - Infections occur within households in discrete time
  - $p$  is the probability an infective infects a susceptible in a household and  $q = 1 - p$  is the chance of escaping infection
  - $N = 334$  families with 3 susceptibles at the beginning
  - Actual chains are not observed
    - \* Just how many are infected at end of epidemic
      - $1 \rightarrow 1 \rightarrow 1$  and  $1 \rightarrow 2$  chain frequency can't be observed!

| Chain                                   | Chain probability | Frequency             | Observed frequency | Final number infected |
|-----------------------------------------|-------------------|-----------------------|--------------------|-----------------------|
| $1 \longrightarrow 0$                   | $q^2$             | $n_1$                 | 34                 | 1                     |
| $1 \longrightarrow 1 \longrightarrow 0$ | $2pq^2$           | $n_{11}$              | 25                 | 2                     |
| $1 \longrightarrow 1 \longrightarrow 1$ | $2p^2q$           | $n_{111}$             | not observed       | 3                     |
| $1 \longrightarrow 2$                   | $p^2$             | $n_{12}$              | not observed       | 3                     |
| <b>Total</b>                            | <b>1</b>          | <b><math>N</math></b> | <b>334</b>         |                       |

- MCMC can augment the data to help infer  $p$

### SIR model

- Initially  $(S(0), I(0), R(0)) = (N - 1, 1, 0)$
- Time infectious:  $T_I \sim (\exp(\gamma))$
- Infectious contacts occur according to Poisson process of rate  $\beta/N$
- Therefore the hazard (Force of Infection) is  $\frac{\beta I(t)}{N}$
- Infection times are not observed therefore estimating parameters difficult
  - Treated as latent variables
  - MCMC augments time to estimate parameters

### Prior distributions

#### Conjugate prior distributions

- Conjugacy
  - Property that the poster distribution follows the same parametric form as the prior distribution
  - Conjugate prior distributions along with appropriate sampling distribution for the data have the same distribution as the posterior distribution
- Beta prior distribution is conjugate family for binomial likelihood
  - Posterior distribution is Beta
- Gamm prior distribution is conjugate family for Poisson likelihood
  - Posterior distribution is Gamma
- Advantage:
  - Computational convenience
  - Can be interpreted as additional data
- Disadvantage:
  - Constrain the form of the prior distribution
- Nonconjugate prior distributions:

- Used when prior knowledge about the distribution of the parameters of interest don't correspond to the conjugate prior distribution
- Computationally more demanding
- Noninformative prior distributions:
  - AKA flat priors
  - Carry little population information
  - Supposed to play a minimal role in the posterior distribution

## Transmission probability

### Data and sampling distribution

- Let's start with a goal of doing inference on the posterior distribution of the transmission probability
- $n$  people exposed once to infection
  - $y$  infected
  - $p$  = transmission probability
  - $q = 1 - p$
- Binomial sampling distribution:

$$\begin{aligned}
 \mathcal{L}(y|p) &= \text{Binomial}(y|n, p) \\
 &= \binom{n}{y} p^y (1-p)^{n-y} \\
 &= \binom{n}{y} p^y q^{n-y}
 \end{aligned}$$

- Given we have a Binomial distribution, we will specify a Beta prior distribution for  $p$

$$\begin{aligned}
 p &\sim \text{Beta}(\alpha, \beta) \\
 \pi(p) &= \text{Beta}(p|\alpha, \beta) \\
 \text{Beta}(p|\alpha, \beta) &= \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1} (1-p)^{\beta-1}, \alpha > 0, \beta > 0 \\
 &\therefore \\
 \text{Beta} &\propto p^{\alpha-1} (1-p)^{\beta-1}
 \end{aligned}$$

$$\begin{aligned}
 \text{Mean: } E(p|\alpha, \beta) &= \frac{\alpha}{\alpha + \beta} \\
 \text{Variance: } \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} &= \frac{E(p|\alpha, \beta)[1 - E(p|\alpha, \beta)]}{\alpha + \beta + 1}
 \end{aligned}$$

- $\alpha > 0, \beta > 0$  indicate “prior sample sizes”
- **Note:**  $\text{Beta} \propto p^{\alpha-1} (1-p)^{\beta-1}$  looks like a binomial distribution!

### Posterior Distribution

$p$

The posterior distribution of the transmission probability

$$\begin{aligned} f(p|y) &\propto \overbrace{p^y(1-p)^{n-y}}^{\text{likelihood}} \overbrace{p^{\alpha-1}(1-p)^{\beta-1}}^{\text{prior}} \\ &= p^{y+\alpha-1}(1-p)^{n-y+\beta-1} \\ &\propto \text{Beta}(p|\alpha+y, \beta+n-y) \end{aligned}$$

- Posterior probability of transmission for a future draw from the population
  - The posterior mean always lies between the prior mean and the sample mean!

$$\begin{aligned} E(p|y) &= \frac{\alpha+y}{\alpha+\beta+n} \\ \text{var}(p|y) &= \frac{E(p|y)[1-E(p|y)]}{\alpha+\beta+n+1} \end{aligned}$$

### Uniform prior distribution

- $[0, 1]$  corresponds to  $\alpha = 1, \beta = 1$ 
  - Essentially no prior on  $p$

$$f(p|y) = \text{Beta}(p|y+1, n-y+1)$$

### Prediction

- Posterior prediction distribution
  - Posterior because conditional on  $y$
  - Predictive because prediction for an observable  $y_{n+1}$

$$\begin{aligned} f(y_{n+1}|y) &= \int f(y_{n+1}, p|y) dp \\ &= \int f(y_{n+1}|p, y) f(p|y) dp \\ &= \int f(y_{n+1}|p) f(p|y) dp \end{aligned}$$

- We can go from line 2 to 3 because  $y$  and  $y_{n+1}$  are conditionally independent given  $p$  in the model